In [1]:
```python
import os
import sys
import pandas as pd
from data_profiling import ProfileReport
```

In [2]:
```python
root_dir = os.path.dirname(os.path.dirname(os.getcwd()))
os.chdir(root_dir)
sys.path.append(os.getcwd())
print(f"Working directory: {os.getcwd()}")
```

```
Working directory: /Users/ramsundar/Documents/IITM-Graded-Projects/Capstone
```

In [3]:
```python
df = pd.read_csv("Database/model_table.csv")
print(f"Loaded {len(df)} rows x {len(df.columns)} columns")
df.head()
```

```
Loaded 25000 rows x 33 columns
```

Out[3]:

	patient_id	age	gender	city	insurance_provider	chronic_flag	registration_date	visit_id	visit_date	department	...	total_billed	avg_payment_days	provider_rejection_rate	doctor_rejectio
0	2964	25	F	Chennai	HealthPlus	1	2025-07-04	3	2025-07-13	ICU	...	44010.80	4.250000	0.149678	0.1
1	4271	42	F	Mumbai	MediCareX	1	2025-10-31	669	2026-01-17	Neurology	...	180737.86	15.222222	0.152480	0.1
2	1826	52	M	Pune	CareOne	1	2025-01-23	1264	2025-10-01	General	...	116149.24	9.500000	0.148655	0.1
3	1224	52	M	Delhi	CareOne	1	2025-06-23	1326	2025-11-05	General	...	93356.58	10.000000	0.148655	0.1
4	1318	39	M	Chennai	HealthPlus	1	2025-11-13	1455	2025-08-20	Cardiology	...	77804.09	20.600000	0.149678	0.1

5 rows × 33 columns

In [4]:
```python
profile = ProfileReport(
    df,
    title="Model Table – Data Quality Report",
    explorative=True,
    progress_bar=False,
)
print("Profile report generated.")
```

```
Profile report generated.
```

In [5]:
```python
output_dir = "Data Quality Report"
os.makedirs(output_dir, exist_ok=True)

html_path = os.path.join(output_dir, "data_quality_report.html")
profile.to_file(html_path)
print(f"HTML report saved to: {html_path}")
```

```
  0%|                                                  | 0/33 [00:00<?, ?it/s]
 21%|██████████                                 | 7/33 [00:00<00:00, 50.43it/s]
```

```
100%|██████████████████████████████████████████████████████████████████████████████| 33/33 [00:00<00:00, 167.52it/s]
HTML report saved to: Data Quality Report/data_quality_report.html
```